

Hybrid-Kinetic PIC Simulation of a Field-Reversed Configuration

Physics, Implementation, Tuning, Limitations, and the Path to 3D HPC Production Runs
Companion documentation for `frc_hybrid_pic.py` (WarpX/PICMI, Ohm's-law hybrid solver)

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1 Purpose and scope

This document describes the physics content, numerical implementation, and practical operation of `frc_hybrid_pic.py`, a hybrid-kinetic particle-in-cell (PIC) simulation of a field-reversed configuration (FRC) plasma built on WarpX 26.05 through the PICMI interface, using WarpX’s Ohm’s-law hybrid solver. The code targets the kinetic regime of compact, high-density FRCs ($n \sim 10^{22} \text{ m}^{-3}$, $T_i \sim \text{keV}$, $r_s \sim 2 \text{ cm}$), where the number of ion gyroradii between the field null and the separatrix is of order unity and finite-Larmor-radius (FLR) physics governs global stability.

The document is organized as follows. Section 2 explains why a hybrid model is the appropriate tool and quantifies the scale separation that makes full explicit PIC intractable. Section 3 states the governing equations exactly as the solver implements them. Section 4 derives the rigid-rotor FRC equilibrium loaded at $t = 0$ and maps each formula to the code. Section 5 covers the derived numerical constraints (grid, timestep, sub-stepping) and the stability conditions behind them. Section 6 documents the Coulomb and D–D fusion modules, including the nuclear-mass bookkeeping that WarpX enforces. Section 7 describes the physics diagnostics. Section 8 is a tuning guide: every adjustable parameter, what it controls physically, and how to move it for better results. Section 9 states the model’s limitations honestly. Section 10 lays out what must be added—physics and software—to run credible 3D simulations of real FRC devices on HPC resources.

2 Why hybrid-kinetic, not full PIC or MHD

2.1 The scale-separation problem

An explicit electromagnetic PIC code must resolve the smallest electron scales: the Debye length λ_D (to avoid finite-grid instability, which numerically heats the plasma) and the electron plasma period ω_{pe}^{-1} (CFL on the light wave and plasma oscillation). For the target FRC parameters the derived scales are:

Quantity	Formula	Value (this run)
Debye length	$\lambda_D = v_{th,e}/\omega_{pe}$	$1.66 \mu\text{m}$
Electron skin depth	$d_e = c/\omega_{pe}$	$53 \mu\text{m}$
Electron gyroradius	$\rho_e = v_{th,e}/\omega_{ce}$	$17 \mu\text{m}$
Ion gyroradius	$\rho_i = v_{th,i}/\omega_{ci}$	2.04 mm
Ion skin depth	$d_i = c/\omega_{pi}$	3.22 mm
Separatrix radius	r_s	20 mm

Table 1: Hierarchy of length scales at $n = 10^{22} \text{ m}^{-3}$, $T_i = 2 \text{ keV}$, $T_e = 0.5 \text{ keV}$, $B_e = 3.17 \text{ T}$. Full PIC must resolve the first row; the hybrid model only the fourth and fifth.

Resolving λ_D over a $2 \times r_s$ domain requires roughly 10^4 cells per dimension and a timestep near 10^{-13} s —about $306\times$ finer in space and three to four orders of magnitude more total work than the hybrid run. The hybrid model removes the electron mass entirely: electrons become a massless, charge-neutralizing fluid, which deletes ω_{pe} , λ_D , and the light wave from the system. The grid then needs only to resolve $\min(d_i, \rho_i)$ and the timestep only the ion gyration ω_{ci}^{-1} , with the dispersive whistler handled by inexpensive magnetic-field sub-stepping. This is the standard approach for kinetic FRC studies (Belova’s HYM, and the WarpX `ohm_solver` module).

2.2 Why not MHD

Single-fluid MHD predicts the FRC to be violently unstable to the internal tilt mode ($n = 1$), in contradiction with experiment. Stability of real FRCs rests on kinetic ion effects—FLR corrections

and the small number of gyroradii in the configuration—quantified by the kinetic parameter

$$\bar{s} \simeq \frac{r_s - R_0}{\rho_i}, \quad (1)$$

where R_0 is the field-null radius. MHD corresponds to $\bar{s} \gg 1$; the kinetic, FLR-stabilized regime is $\bar{s} \sim \mathcal{O}(1)$. The present run has $\bar{s} = 2.9$, squarely in the regime where ion kinetics matter and a kinetic-ion model is mandatory. The two-fluid (Hall) terms retained in the hybrid Ohm’s law additionally introduce the ion-scale physics—whistler dynamics, Hall reconnection, ion Bernstein waves—that single-fluid models cannot represent.

2.3 Emergent kinetic physics

A point of philosophy that shapes the code: ion Landau damping and ion Bernstein waves are not modules to be switched on. With kinetic ions they appear automatically, *provided* the grid resolves the relevant scales ($\Delta x \lesssim \rho_i$ for Bernstein structure) and the particle count keeps velocity-space noise below the signal ($N_{\text{ppc}} \gtrsim 100\text{--}400$). What the code adds is (a) resolution that admits these effects and (b) diagnostics that measure them (Sec. 7).

3 The hybrid-kinetic model

3.1 Kinetic ions

Deuterium ions are evolved as macroparticles with the standard Boris push,

$$m_i \frac{d\mathbf{v}_p}{dt} = q_i (\mathbf{E} + \mathbf{v}_p \times \mathbf{B}), \quad \frac{d\mathbf{x}_p}{dt} = \mathbf{v}_p, \quad (2)$$

with cubic (shape-factor 3) interpolation between particles and grid. Momentum inputs throughout the code are *proper velocities* $u = \gamma v$ in m/s (for $v \ll c$, simply v in m/s)—never mv . The ion charge density ρ_i and current $\mathbf{J}_i$ are deposited on the grid each step.

3.2 Massless electron fluid and the generalized Ohm’s law

Quasineutrality ($n_e = n_i \equiv n$) plus the massless-electron momentum equation yield the electric field algebraically:

$$\boxed{\mathbf{E} = -\mathbf{u}_i \times \mathbf{B} + \underbrace{\frac{\mathbf{J} \times \mathbf{B}}{en}}_{\text{Hall}} - \underbrace{\frac{\nabla p_e}{en}}_{\text{e}^- \text{ pressure}} + \eta \mathbf{J} - \eta_H \nabla^2 \mathbf{J}} \quad (3)$$

where $\mathbf{u}_i$ is the ion bulk velocity from particle moments. The total current comes from Ampère’s law *without displacement current*,

$$\mathbf{J} = \frac{1}{\mu_0} \nabla \times \mathbf{B}, \quad \mathbf{J}_e = \mathbf{J} - \mathbf{J}_i, \quad (4)$$

which is what removes the light wave and ω_{pe} from the system. The magnetic field is advanced with Faraday’s law,

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}, \quad (5)$$

sub-cycled M times per ion step (CFG.SUBSTEPS) to resolve the grid-scale whistler (Sec. 5.4).

The Hall term is what makes this a *two-fluid* model: below d_i the magnetic field is frozen into the electron fluid rather than the bulk plasma, enabling fast Hall reconnection and supporting whistler and ion-cyclotron-branch waves. WarpX’s hybrid solver hard-wires the Hall and electron-pressure terms (a literal Hall-coefficient scan would require a small source patch); the physically equivalent scan is over $\bar{s}$, since smaller $\bar{s}$ strengthens two-fluid/FLR effects relative to MHD.

3.3 Electron closure

The electron fluid is closed polytropically,

$$p_e = n_0 k_B T_{e0} \left(\frac{n}{n_0} \right)^\gamma, \quad \gamma = \frac{5}{3} \text{ (adiabatic; } \gamma = 1 \text{ for isothermal)}, \quad (6)$$

with reference density $n_0 = \text{CFG.N_NULL}$ and electron temperature $T_{e0} = \text{CFG.T_E}$. There is no electron energy equation: electron heating by reconnection or compression is not captured (Sec. 9).

3.4 Resistivity as physics, not fudge

The scalar resistivity η (`CFG.ETA`) plays a dual role. Numerically it prevents current sheets (e.g. the merging X-point) from collapsing below the grid scale. Physically it *sets the reconnection rate* in the resistive layer and hence the merging/heating timeline; it must therefore be reported with any result. Spitzer resistivity at keV temperatures,

$$\eta_{\text{Sp}} \approx 5.2 \times 10^{-5} \frac{Z \ln \Lambda}{T_e^{3/2} [\text{eV}]} \Omega \text{ m}, \quad (7)$$

is of order $10^{-7} \Omega \text{ m}$ here—far below the anomalous value $\eta = 10^{-4} \Omega \text{ m}$ used by default, which stands in for under-resolved sheet microphysics. The hyper-resistivity η_H (`CFG.ETA_HYPER`) damps only grid-scale current ripples and should be kept as small as stability allows.

3.5 The vacuum problem and the background halo

Equation (3) divides by n in the Hall and pressure terms: a hybrid model *cannot tolerate vacuum*. Two coordinated floors enforce this: the loaded density profile is floored at a halo fraction $n_{\text{bg}} = f_{\text{bg}} n_0$ with $f_{\text{bg}} = \text{CFG.N_BG_FRAC} = 0.05$, and the solver applies a matching internal floor (`n_floor`). The halo is physically motivated—real FRCs are surrounded by open-field-line plasma at a few percent of peak density—but it is also a numerical necessity: with $f_{\text{bg}} \rightarrow 0$ the local $d_i \propto n^{-1/2}$ diverges and the grid-scale whistler frequency with it, producing non-finite fields on the first solve (observed empirically as the step-1 NaN abort).

4 The rigid-rotor FRC equilibrium

4.1 Magnetic profile

The initial state is a pressure-balanced rigid-rotor (RR) profile. The axial field, implemented in `bz_profile_expr()`, is

$$B_z(r) = B_e \tanh \left(\frac{(r/R_0)^2 - 1}{\delta} \right), \quad (8)$$

with B_e the external field, R_0 the field-null radius, and $\delta = \text{CFG.FIELD_SHARP}$ the dimensionless current-sheet width. The field is reversed ($B_z < 0$) inside R_0 , vanishes at the null, and asymptotes to $+B_e$ outside. For the idealized RR equilibrium the closed-flux condition fixes

$$R_0 = \frac{r_s}{\sqrt{2}}, \quad (9)$$

set in `derive_parameters()` (not hand-editable).

4.2 Pressure balance fixes everything else

Radial pressure balance,

$$p(r) + \frac{B_z^2(r)}{2\mu_0} = \frac{B_e^2}{2\mu_0}, \quad (10)$$

evaluated at the null where $B_z = 0$ and $\beta = 1$, determines the external field from the chosen density and temperatures:

$$\boxed{B_e = \sqrt{2\mu_0 n_0 k_B (T_i + T_e)}} \Rightarrow B_e = 3.17 \text{ T here.} \quad (11)$$

B_e is therefore *not a free parameter*; editing it independently of (n_0, T_i, T_e) destroys force balance and produces a violent initial transient. The density profile follows from Eq. (10) with $p = nk_B(T_i + T_e)$:

$$n(r) = n_0 \max \left[f_{\text{bg}}, 1 - \left(\frac{B_z(r)}{B_e} \right)^2 \right], \quad (12)$$

peaking at the null and falling to the halo floor where $|B_z| \rightarrow B_e$ (`density_profile_expr()`). A useful exact FRC relation worth checking in post-processing is the average-beta theorem, $\langle \beta \rangle = 1 - \frac{1}{2} x_s^2$ with $x_s = r_s/r_w$ the ratio of separatrix to wall radius.

4.3 Diamagnetic ion flow

The equilibrium current that sustains the field reversal is loaded as an azimuthal ion drift. From Ampère's law at the midplane, $\mu_0 J_\phi = -\partial B_z / \partial r$, and $J_\phi = enu_\phi$:

$$u_\phi(r) = -\frac{1}{\mu_0 e n(r)} \frac{\partial B_z}{\partial r}, \quad \frac{\partial B_z}{\partial r} = B_e \operatorname{sech}^2 \left(\frac{(r/R_0)^2 - 1}{\delta} \right) \frac{2r}{R_0^2 \delta}, \quad (13)$$

implemented in `diamagnetic_uphi_expr()` with the denominator floored at the same halo density as Eq. (12). The full diamagnetic current is placed on the ions; the electron fluid carries whatever residual current Eq. (4) demands, and the configuration relaxes self-consistently over the first few gyroperiods. Note again that u_ϕ enters WarpX as a proper velocity in m/s.

4.4 Geometry, elongation, and scenarios

The separatrix half-length is $Z_s = E r_s$ with elongation $E = \text{CFG.ELONGATION}$; the computational domain spans $1.3 r_s$ radially and $2Z_s$ axially. Three scenarios are supported by `CFG.SCENARIO`: "equilibrium" (single FRC relaxation and confinement), "merging" (two RR plasmoids at $z = \pm z_0$ counter-streaming at Mach $M_A = \text{CFG.MERGE_MACH}$ in units of v_A , driving X-point reconnection and ion heating), and "compression" (a hook for ramping B_e by `CFG.COMPRESS_RATIO` over ~ 20 gyroperiods, giving adiabatic heating with $T_\perp \propto B$ from μ -invariance).

5 Numerical design: every constraint derived

5.1 Characteristic frequencies and speeds

`derive_parameters()` computes (SI, $v_{\text{th}} = \sqrt{k_B T/m}$):

$$\omega_{pe} = \sqrt{\frac{ne^2}{\epsilon_0 m_e}}, \quad \omega_{pi} = \sqrt{\frac{ne^2}{\epsilon_0 m_i}}, \quad \omega_{ce} = \frac{eB_e}{m_e}, \quad \omega_{ci} = \frac{eB_e}{m_i}, \quad (14)$$

$$\lambda_D = \frac{v_{\text{th},e}}{\omega_{pe}}, \quad \rho_i = \frac{v_{\text{th},i}}{\omega_{ci}}, \quad d_i = \frac{c}{\omega_{pi}}, \quad v_A = \frac{B_e}{\sqrt{\mu_0 n m_i}}. \quad (15)$$

5.2 Grid

The cell size resolves the smaller of the two ion scales,

$$\Delta x = \frac{\min(d_i, \rho_i)}{N_{\text{cpd}}}, \quad N_{\text{cpd}} = \text{CFG.CELLS_PER_DI} = 4, \quad (16)$$

giving $\Delta x \approx 0.51$ mm, i.e. $\Delta x/\rho_i = 0.25$ (adequate for FLR and the first few Bernstein harmonics) and $\Delta x/d_i = 0.16$. Cell counts are rounded *up* to multiples of the AMReX `blocking_factor` (8), yielding 56×320 cells in RZ; AMReX aborts otherwise. The $r = 0$ boundary is the coordinate axis, not a wall: both field and particle boundaries there must be `none` (WarpX asserts this).

5.3 Timestep

The ion gyration sets the step,

$$\Delta t = \frac{f_{\omega_{ci}}}{\omega_{ci}}, \quad f_{\omega_{ci}} = \text{CFG.DT_WCI} = 0.05 \Rightarrow \Delta t = 3.29 \times 10^{-10} \text{ s}, \quad (17)$$

about 126 steps per gyroperiod ($\omega_{ci}\Delta t = 0.05 \ll 0.25$). Because the hybrid model has no light wave, WarpX cannot auto-set a CFL timestep: a constant Δt *must* be supplied (`warpX.const_dt`, passed as `time_step_size` to `picmi.Simulation`).

5.4 Whistler sub-stepping

In the hybrid model the whistler branch does not saturate at ω_{ce} (no electron inertia): at grid scale,

$$\omega_w \approx (k d_i)^2 \omega_{ci}, \quad k_{\text{max}} = \frac{\pi}{\Delta x} \Rightarrow k_{\text{max}} d_i \approx 20, \quad \omega_w \approx 400 \omega_{ci}, \quad (18)$$

so one full ion step accumulates $\omega_w \Delta t \approx 20$ radians of whistler phase—and more in the halo, where $d_i \propto n^{-1/2}$ grows. The field advance Eq. (5) is therefore sub-cycled $M = \text{CFG.SUBSTEPS} = 250$ times per step, keeping the per-substep phase below ~ 0.4 rad domain-wide. The empirical signature of insufficient M is a non-finite **E** abort on the first steps. The stability margin can be traded: halving `DT_WCI` permits roughly half the substeps at fixed cost.

5.5 Particle statistics

Macro-ion count per cell, $N_{\text{ppc}} = \text{CFG.PPC} = 200$. Velocity-space noise scales as $N_{\text{ppc}}^{-1/2}$; the Landau-decay and Bernstein-spectrum diagnostics need $N_{\text{ppc}} \gtrsim 100$ –400 to keep mode amplitudes above the noise floor. Total macro-ions here: $56 \times 320 \times 200 \approx 3.6$ M.

6 Collisions and D–D fusion

6.1 Coulomb collisions

Intra-species ion–ion Coulomb scattering uses WarpX’s binary-pairing module (Nambu/Pérez-type), registered through PICMI with a super-cycling interval (`ndt.supercycle` = 2: collide every second step). At these parameters the ion–ion collision time exceeds the run duration, so Coulomb scattering is a slow thermalizing correction, not a dominant process; it matters most for relaxing the loaded distribution and for slowing fusion products if those are later made collisional. **Compatibility constraint:** WarpX’s binary-collision framework (Coulomb *and* fusion) requires $n_{\text{rz modes}} = 1$ in RZ; the code enforces this automatically whenever collisions are enabled.

6.2 D–D fusion: both branches

Fusion is implemented by WarpX’s `nuclearfusion` binary-collision type, registered natively (the PICMI layer has no fusion class) as two blocks:

$$\text{D} + \text{D} \rightarrow \text{He}^3 (0.82 \text{ MeV}) + n (2.45 \text{ MeV}), \quad Q = 3.27 \text{ MeV}, \quad (19)$$

$$\text{D} + \text{D} \rightarrow \text{T} (1.01 \text{ MeV}) + p (3.02 \text{ MeV}), \quad Q = 4.03 \text{ MeV}, \quad (20)$$

each $\approx 50\%$ branching at keV energies. The cross-sections are WarpX’s built-in (Bosch–Hale-class) parameterizations. Each species must carry the native input `species_type` (deuterium, helium3, tritium, proton, neutron): the fusion module classifies reactants by type, not by name or mass.

Nuclear masses, not atomic. WarpX validates the kinematics of every fusion event against the species mass deficit, $Q = (\sum m_{\text{react}} - \sum m_{\text{prod}})c^2$, to a relative tolerance of 10^{-2} . The masses supplied must therefore be *bare-nucleus* values (deuteron 2.013553 u, helion 3.014932 u, triton 3.015501 u, proton 1.007276 u, neutron 1.008665 u). Using atomic masses unbalances the electron count between the two sides—for $\text{D}+\text{D}\rightarrow\text{T}+p$, two bound electrons in versus one out—inflating Q by exactly $m_e c^2 = 511 \text{ keV}$ and triggering an abort. This is a sharp, instructive failure mode and is documented in the code header.

Rare-event sampling. Thermonuclear D–D rates at 2 keV are tiny; without enhancement no macro-fusion would occur in 10^4 steps. The `event_multiplier` (`CFG.FUSION_MULT = 10^6`) boosts the per-pair reaction probability and creates product macroparticles with correspondingly reduced weight, leaving the *physical* reaction rate unbiased while making the statistics measurable. The companion knobs `probability_threshold` and `probability_target_value` guard against per-pair probabilities approaching unity; if WarpX warns about saturated probabilities, reduce the multiplier.

7 Diagnostics

The `Diagnostics` callback (installed after each step; the instance carries a `__name__` because WarpX’s callback registry requires one) records:

Energy partition. Magnetic energy $\int B^2/2\mu_0 dV$ and total ion kinetic energy each `DIAG_EVERY` steps. In merging runs, the signature of reconnection is magnetic energy converting to ion kinetic/thermal energy; in all runs, secular drift of the total is the primary numerical-health monitor.

Ion temperature. From the velocity variance about the mean flow, $k_B T_i = m_i \langle \delta v^2 \rangle / 3$ (flow-corrected, so the diamagnetic drift is not miscounted as heat). This is the ion-heating diagnostic for merging and compression.

Poloidal flux and reconnection. The axisymmetric flux

$$\psi(r, z) = \int_0^r B_z(r', z) 2\pi r' dr' \quad (21)$$

is accumulated by radial quadrature; in merging runs the midplane (X-point) flux history measures the reconnected flux and hence the reconnection rate, which should be compared against the Hall-reconnection expectation ($\sim 0.1 B_e v_A$ inflow-normalized) and its dependence on η .

Confinement times. Counting macro-ions and their thermal energy inside the separatrix ($r < r_s$, $|z| < Z_s$) gives

$$\tau_N = -\frac{N}{dN/dt}, \quad \tau_E = -\frac{W}{dW/dt}, \quad (22)$$

evaluated in post-processing. With the 5% halo present, the counts must include only particles that *started* inside the separatrix (tag at $t = 0$), or halo particles drifting through will contaminate the loss rate.

Ion Bernstein waves. A fixed radial probe records $E_r(t)$; its power spectrum exhibits peaks at the cyclotron harmonics $\omega \approx n \omega_{ci}$ characteristic of IBW. A full ω - k dispersion map requires storing E_r along a radial line each diagnostic interval and 2D-FFTING in (r, t) —noted in the code as the natural upgrade.

Landau damping. Seeding a low-amplitude ion-acoustic or drift perturbation and fitting the exponential decay of the mode amplitude yields the kinetic damping rate; with $T_e/T_i = 0.25$ here, ion Landau damping of ion-acoustic fluctuations is strong, and the measured decay should reflect that.

8 Tuning guide: what to turn, and what it does

Table 2 summarizes every operationally relevant knob. The discussion below adds the physics reasoning.

Recommended convergence protocol. (1) Establish a stable baseline at the defaults with `MAX_STEPS` ~ 2000 (a few gyroperiods) and confirm energy conservation to $\lesssim 1$ –2%. (2) Double PPC and confirm diagnostics move less than the quantity of interest. (3) Increase `CELLS_PER_DI` to 6 and repeat. (4) For merging: scan `ETA` a decade either way and present the reconnected flux versus η —the result is kinetic only where the dependence flattens. (5) Only then commit to long (2×10^4 -step) production runs.

9 Limitations

9.1 Geometry: what RZ can never show

The default RZ run is strictly axisymmetric (and with collisions enabled, WarpX restricts RZ to a single azimuthal mode). The two instabilities that historically define FRC viability—the internal **tilt** ($n = 1$) and the **rotational** ($n = 2$) modes—break axisymmetry and *cannot appear* in this geometry; no post-processing recovers them. RZ answers questions about equilibrium, merging/reconnection, compression heating, axisymmetric confinement, and radial wave structure. Stability questions require the 3D path (Sec. 10).

9.2 Electron physics

Massless electrons remove electron Landau damping, electron heating, electron pressure anisotropy, and the d_e -scale physics of the reconnection diffusion region; the whistler branch is unphysically undamped at small scales (it is controlled numerically by η_H and sub-stepping rather than by electron inertia). The polytropic closure carries no electron heat flux and no electron-energy evolution: T_e is effectively prescribed. Any conclusion sensitive to electron energetics (radiative balance, sheath physics, reconnection electron heating) is outside this model.

Parameter	Default	Effect / guidance
N_NULL, T_I, T_E, R_SEP	10^{22} , 2000, 500 eV, 2 cm	Define the device. B_e , R_0 , all scales follow. Changing any re-derives the whole run; never hand-edit B_e .
ELONGATION	4.0	Z_s/r_s . Prolate ($E \gg 1$) FRCs are tilt-prone in MHD; oblate ($E \lesssim 1$) tilt-stable. Also sets domain length and cost $\propto E$.
FIELD_SHARP (δ)	0.5	Current-sheet width. Smaller δ = thinner, more realistic sheet but needs finer grid; keep sheet width $\gtrsim 4\Delta x$.
CELLS_PER_DI	4	Resolution of $\min(d_i, \rho_i)$. Raise to 6–8 for clean Bernstein harmonics $n \gtrsim 3$ and sharper sheets; cost $\propto N_{\text{cpd}}^2$ (RZ) or 3 (3D).
PPC	200	Velocity-space noise $\propto N_{\text{ppc}}^{-1/2}$. 100 is a floor for kinetic diagnostics; 400+ for quantitative Landau rates. Linear cost.
DT_WCI	0.05	Gyro-resolution. 0.02–0.10 sensible; smaller also relaxes the substep demand.
SUBSTEPS	250	Whistler CFL, Eq. (18). Too few $\Rightarrow$ non-finite E within steps; find the stability edge by descending (250 $\rightarrow$ 150 $\rightarrow$ 100) once a run survives. Scale up if CELLS_PER_DI or the density floor changes.
ETA	$10^{-4} \Omega m$	<i>Sets the reconnection rate</i> and sheet thickness. Scan it (10^{-5} – 10^{-3}) in merging studies and report it; results that depend strongly on η are resistive, not kinetic.
ETA_HYPER	10^{-10}	Grid-scale smoothing only. Increase one decade at a time if grid-ripple noise in J appears; too large damps real Bernstein/whistler structure.
N_BG_FRAC	0.05	Halo density. Lower is more “vacuum-like” but raises the local whistler frequency ($\propto 1/n$) demanding more substeps; below ~ 0.01 expect instability. Affects edge physics and τ_N bookkeeping.
E_GAMMA	5/3	Electron closure. $\gamma = 1$ (isothermal) is more forgiving and a useful sensitivity check on any electron-pressure-driven result.
MERGE_MACH	0.5	Collision drive. Higher Mach $\Rightarrow$ faster, more violent reconnection and stronger ion heating; supersonic merging (> 1) generates shocks.
FUSION_MULT	10^6	Product statistics. Raise until product counts are smooth; lower if WarpX warns about reaction probabilities near 1 (biased kinematics).
TILT_PERT	0.02	3D only: $n = 1$ seed amplitude. Must exceed PIC noise but stay linear ($\lesssim$ few %) to measure a growth rate cleanly.

Table 2: Tunable parameters, defaults, and guidance.

9.3 Reconnection-rate dependence on η

With a scalar resistivity, the merging X-point dissipation is parameterized, not derived. The Hall term gives qualitatively correct fast reconnection, but the quantitative rate retains η -dependence until the sheet’s kinetic scales are resolved and an electron model with inertia or a kinetic closure is used.

9.4 Equilibrium idealization

The rigid-rotor profile is a one-dimensional pressure-balance solution, not a solution of the Grad–Shafranov equation in the actual finite-elongation domain; the configuration therefore rings and relaxes for the first few gyroperiods, and quantitative comparisons should start after this transient. The halo ($f_{\text{bg}} = 5\%$) is partly numerical; edge and open-field-line physics (end loss along open lines, mirror dynamics in the cusp regions) are not represented—the axial boundaries simply absorb particles.

9.5 Missing physics inventory

No neutrals, ionization, or charge exchange; no impurities or radiation (bremsstrahlung, line); no RF or neutral-beam sources; no rotating-magnetic-field (RMF) or other current drive; no wall/sheath model; fusion products are created with correct kinematics but the run duration ($\sim \mu\text{s}$) is far below their slowing-down time, so alpha-channeling-like product physics is out of scope. Statistical limits: with $N_{\text{ppc}} = 200$, weak kinetic signals (high Bernstein harmonics, slow Landau rates) sit near the noise floor.

10 The path to 3D and real-device simulations on HPC

10.1 Cost of going 3D at present parameters

The code already contains the 3D branch (`CFG.GEOMETRY="3D"`), which is *required* for tilt and rotational physics. At the current resolution the Cartesian domain is $\sim 112 \times 112 \times 320 \approx 4 \times 10^6$ cells; at 200 PPC that is $\sim 8 \times 10^8$ macro-ions, ~ 100 GB of particle memory, and $\sim 1.6 \times 10^{13}$ particle-pushes for a 2×10^4 -step run—a multi-GPU job, comfortably within a handful of A100/H100-class nodes for WarpX, whose GPU port routinely sustains 10^8 – 10^9 pushes/s per GPU. The field sub-cycling (250 substeps over 4×10^6 cells) adds a comparable field-solve load; both scale well under MPI domain decomposition with AMReX load balancing. Practical first 3D campaign: hold RZ-validated parameters, seed TILT.PERT, run ~ 5000 steps, and measure the $n = 1$ growth rate versus $\bar{s}$ and E —the classic kinetic-FRC stability scan.

10.2 Scaling to a real device

Real FRCs (e.g. C-2W-class: $r_s \sim 0.4$ m, $n \sim 3 \times 10^{19} \text{ m}^{-3}$, $T_i \sim 1$ keV, $B_e \sim 0.1$ T) have larger ρ_i (cm-scale) but vastly larger volume: the cell count scales as $(r_s/\rho_i)^2 \times (Z_s/\rho_i)$ and the step count as the physics duration over ω_{ci}^{-1} . Confinement-relevant durations (ms) at $\omega_{ci} \sim 10^7 \text{ s}^{-1}$ mean 10^5 – 10^6 ion steps. This is leadership-class territory (10^2 – 10^4 GPU-node-hours per run) and motivates every item below.

10.3 Physics additions required

Grad–Shafranov equilibrium initialization. Replace the rigid-rotor load with a numerical solution of

$$\Delta^* \psi \equiv r \frac{\partial}{\partial r} \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{\partial^2 \psi}{\partial z^2} = -\mu_0 r^2 \frac{dp}{d\psi}, \quad (23)$$

in the actual elongated domain, eliminating the startup transient. A physics-informed neural network or standard elliptic solver for $\psi(r, z)$ feeding the particle loader is a natural pre-processing module, and directly reuses an existing Grad–Shafranov PINN workflow.

Electron model upgrades. In rough order of value: (i) electron energy equation (evolve T_e with ohmic heating ηJ^2 and compressional work), (ii) finite electron inertia in Ohm’s law (restores d_e -scale physics and caps the whistler at ω_{ce} , *reducing* the substep burden), (iii) Landau-fluid or gyro-fluid closures for electron heat flux, (iv) anisotropic ($p_{e\parallel}, p_{e\perp}$) pressure.

Device-relevant sources and sinks. Formation and sustainment physics: theta-pinch formation fields and translation (time-dependent boundary coils), neutral-beam fast-ion species (kinetic ions with slowing-down source—directly relevant to beam-driven FRCs), rotating-magnetic-field current drive (time-dependent applied $B_{\perp}(t)$ at the boundary), end-loss regions with mirror coils and realistic open-line boundary conditions, and a neutral/charge-exchange module for edge fueling. Radiation (bremsstrahlung at minimum) closes the power balance.

Fusion-product transport. For burn-relevant questions, retain $\text{He}^3/\text{T}/\text{p}$ as kinetic species with Coulomb slowing on the bulk, enabling product confinement and pressure-contribution studies—the RZ orbit physics of MeV products in an FRC (betatron orbits, loss cones) is itself a publishable diagnostic.

10.4 Software and workflow additions

Production 3D runs need: openPMD/HDF5 field and particle output with in-situ reduction (the current in-memory Python diagnostics do not scale across ranks and must become reduced diagnostics computed per-rank and MPI-reduced); checkpoint/restart for queue-window resilience; AMReX dynamic load balancing (particle-weighted) since the FRC concentrates particles near the null; azimuthal mode-amplitude diagnostics ($n = 1, 2, 3$ Fourier amplitudes of B_r at the null radius) as the primary 3D stability observable; parameter-scan orchestration ($\bar{s}$, E , η , M_A grids); and verification milestones—reproduce the WarpX `ohm_solver` regression examples, then an established kinetic-FRC benchmark (tilt growth-rate versus $\bar{s}$ against published HYM results) before producing new science. Mesh refinement around the X-point during merging is the single highest-leverage solver feature for reconnection studies.

11 Summary

`frc_hybrid_pic.py` implements a physically self-consistent, fully-derived hybrid-kinetic FRC simulation: a $\beta = 1$ rigid-rotor equilibrium whose every parameter follows from pressure balance; kinetic deuterium ions with the full diamagnetic current; a massless-electron generalized Ohm’s law with Hall physics, polytropic closure, and explicitly acknowledged resistive dissipation; binary Coulomb collisions and both D–D fusion branches with rigorous nuclear-mass kinematics; and diagnostics aimed at the quantities that make a run science rather than animation—energy conservation, ion heating, reconnected flux, confinement times, and emergent kinetic wave activity. Its honest domain of validity is axisymmetric, ion-kinetic FRC physics on microsecond timescales; its clearly marked frontier—3D stability, electron energetics, device-scale duration—is reachable with the HPC roadmap of Sec. 10 on the same WarpX foundation.